Probing Long-Range Interactions in the Ising Model Using Bootstrapping Methods

P. Todd and A. (C.) Yonter. Supervised by Prof. S. Caron-Huot

McGill University, Department of Physics, 3600 University Street, Montréal, Quebec H3A2T8 Canada

Overview

We study the $d = 2$ **Ising model** with no external magnetic field on the $\mathbb{Z}^2$ lattice, using a **bootstrap approach** which follows the framework of [1]. Why?

- For $d > 2$, the Ising model admits no exact solution, and the bootstrap may sometimes be more precise than Monte-Carlo simulations.
- The Ising model is a good testing ground for bootstrapping techniques applied to more complicated theories, such as lattice Yang-Mills theory.

We obtain results for some short-range correlators, including those already found in [1], with a particular attention toward how the sizes of the constraints affect the resulting bounds. We also give an approximation scheme which reduces the size of computational space to be linear in the length scale, which may permit the bootstrap for slightly longer-range correlators (~ 10 sites).

What is a bootstrap?

A non-perturbative method which uses consistency, symmetry, and algebraic constraints to constrain the observables of a theory without needing a microscopic Lagrangian or Hamiltonian. **Example**: $d = 0$ ϕ^4 theory, where $\mathcal{S}(x) = \frac{x^2}{2} + \frac{\lambda}{4}x^4$.

$$W_k = \int dx x^k e^{-\mathcal{S}(x)} \implies (k+1)W_k = W_{k+2} + \lambda W_{k+4} \quad (*)$$

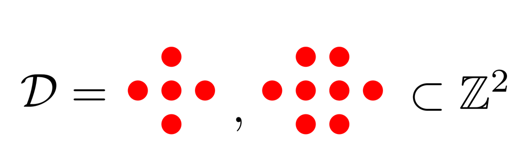
$$\int_{-\infty}^{\infty} dx (\alpha_i x^i)^2 e^{-\mathcal{S}(x)} \geq 0 \iff \begin{pmatrix} W_0 & W_1 & \cdots \\ W_1 & W_2 & \cdots \\ \vdots & \vdots & \ddots \end{pmatrix} \succeq 0 \quad (\dagger)$$

Write $W_k = W_k(W_2)$ using $(*)$ and feed the positive semidefinite (PSD) matrix $(\dagger)$ into a semidefinite programming solver to compute bounds on W_2. **These bounds will be rigorous.**

Our program, based on [1]

Below is a basic description of our program that computes bounds on a given spin correlator $\langle \underline{s}_A \rangle$ for some reasonably small $A \subset \mathbb{Z}^2$:

Pick $\mathcal{D} \subset \mathbb{Z}^2$ with $A \subseteq \mathcal{D}$ and only allow interactions on $\mathcal{D}$.



Compute the analog of $(*)$ and reduce to a set of free correlators, including $\langle s \rangle$ → Compute spin-flip equations (§1)

Compute the analog of $(\dagger)$ which contains $\langle s \rangle$, mostly from reflections of A. → Compute PSD matrices (§2)

Pass these matrices into `SDPB`, an arbitrary-precision semidefinite program solver [3]. → `SDPB` (§3)

Upper and lower bounds on $\langle s \rangle$! → $\langle s \rangle_{\min}$, $\langle s \rangle_{\max}$

References

[1] Minjae Cho, Barak Gabai, Ying-Hsuan Lin, Victor A. Rodriguez, Joshua Sandor, and Xi Yin, *Bootstrapping the ising model on the lattice*, 2022.

[2] Vladimir Kazakov and Zechuan Zheng, *Bootstrap for finite N lattice yang-mills theory*, 2024

[3] David S. D., *Sdpb: A semi-definite programming solver for conformal bootstrap*, 2018.

Acknowledgements

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1. Computing spin-flip equations

For a given $A \subset \mathbb{Z}^2$, one can flip $s_z \mapsto -s_z$ for any $z \in \mathbb{Z}^2$ to obtain a **spin-flip equation**:

$$2\chi_A(z) \langle \underline{s}_A \rangle + \sum_{i=1}^{4} \alpha_i \langle \underline{s}_A w_z^i \rangle - \beta_i \langle \underline{s}_A s_z w_z^i \rangle = 0,$$

where α_i, β_i are functions of the coupling J. We enumerate all spin-flip equations with $(A, z) \in \mathcal{D} \times \mathcal{D}^o$ that are unrelated by symmetries of $\mathbb{Z}^2$ (rotations, reflections and translations), and collect them into a matrix system that is solved numerically.

Domain $\mathcal{D}$	$A \subset \mathcal{D}$	ind. spin-flip equations	ind. $\langle s \rangle$
131:	6	2	4
1331:	36	28	8
$13^3 1$:	245	225	20
$13^4 1$:	1914	1838	76

2. Computing PSD matrices

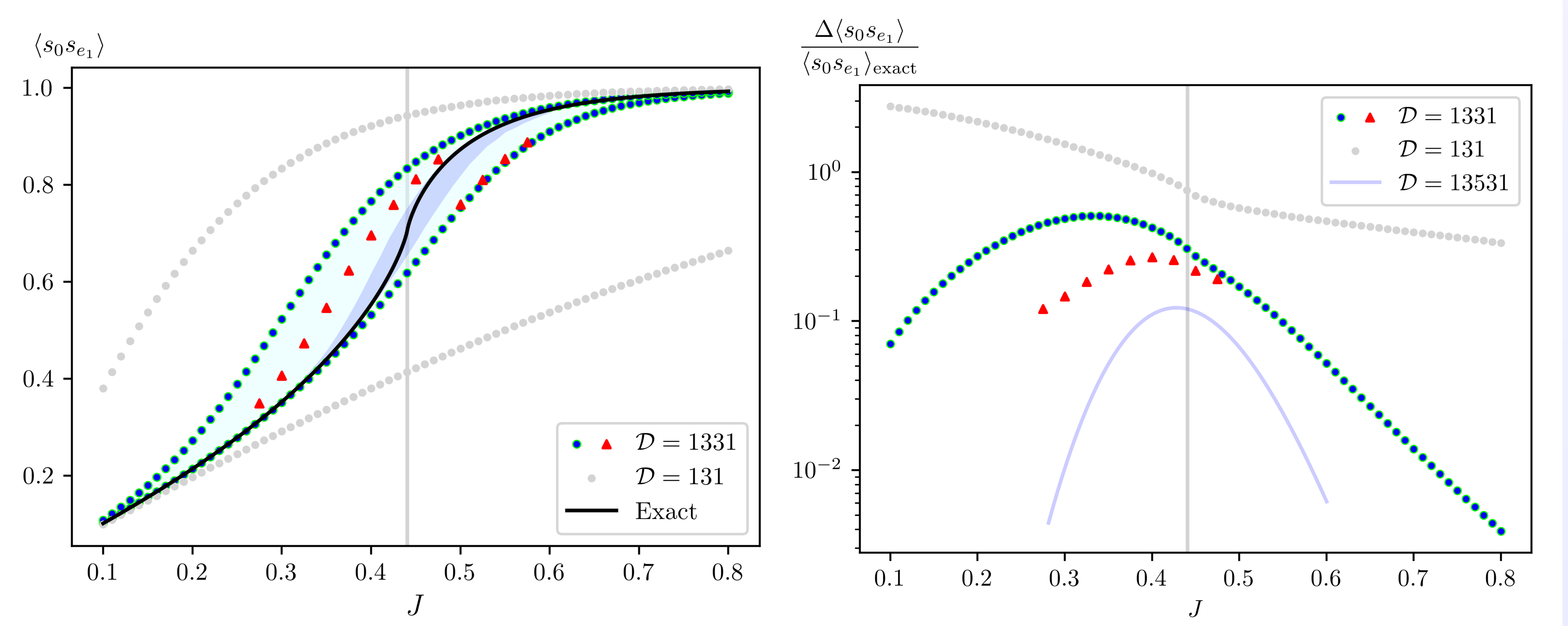
Many positivity constraints are applicable to the Ising model on the lattice, some three of which can be written as the positive semi-definiteness (positivity of all eigenvalues) of matrices. For any $A \subseteq \mathcal{D}$, write $\langle \underline{s}_A \rangle = \langle \prod_{x \in A} s_x \rangle$ and $S_i(B)$ the i'th subset of $B \subset \mathbb{Z}^2$, then the positivity conditions are:

- Square positivity: $\left(\left\langle \underline{s}_{S_i(\mathcal{D})} \underline{s}_{S_j(\mathcal{D})} \right\rangle \right) \succeq 0$,
- Griffiths positivity: $\left\langle \underline{s}_{S_i(\mathcal{D})} \right\rangle \succeq 0$ (or equivalently ≥ 0 here) $\forall i$,
- Reflection positivity: $\left(\left\langle \underline{s}_{S_i(H)} \underline{s}_{S_j(R(H))} \right\rangle \right) \succeq 0$ $\forall v \in \mathbb{Z}^2$, $c \in \mathbb{Z}$ s.t. $H = \{x \in \mathcal{D} | x \cdot v \geq c\} \neq \{\}$, $R(H) \subseteq \mathcal{D}$, and $R(\mathbb{Z}^2) = \mathbb{Z}^2$ for R the reflection about the plane with normal v and offset c from the origin. In 2D, the allowed reflections are vertical and horizontal with $c \in \frac{1}{2}\mathbb{Z}$, and diagonal with $c \in \mathbb{Z}$.

We then express each entry of each matrix in terms of the independent $\langle s \rangle$, and those matrices are then precisely what goes into `SDPB`!

3. `SDPB` and bootstrap calculations

Below is a plot of the bounds obtained on the spin correlator $\langle s_0 s_{e_1} \rangle$ for $\mathcal{D} = 131$ and $\mathcal{D} = 1331$ (left), as well as a plot of their width (right). For blue 1331 data, only matrices up to size 8×8 were used, and the red 1331 data uses all matrices, the largest of which was 41×41. These larger matrices took significantly more time, hence the trends are less resolved.

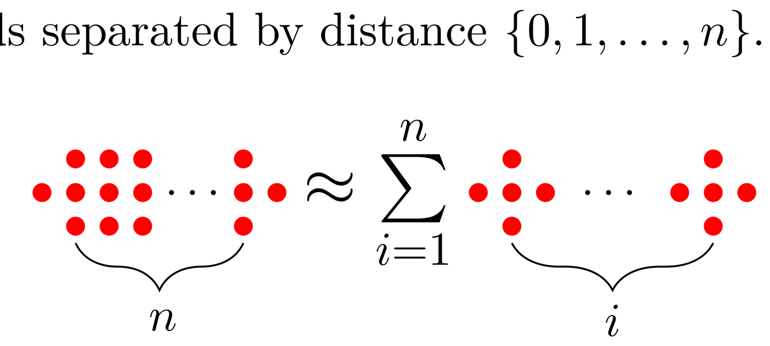


Takeaway 1: $\mathcal{D} = 131$ gives weak bounds (already known), but $\mathcal{D} = 1331$ is strong enough to observe the trend of the correlator as a function of J (solid black line). This suggests that the reason $\mathcal{D} = 131$ is poor may also be because $\langle s_0 s_{e_1} \rangle$ cannot be entirely contained in $\mathcal{D}^o$, not only because it has very few sites.

Takeaway 2: The larger matrices only play a significant role in the upper bound; the lower bound is unchanged up to 4 decimal places. This is likely related to the geometry of the exact trend, as the bounds appear to always follow continuous, smooth lines.

Approximation scheme for studying long-range correlators

The choice of $\mathcal{D} \subset \mathbb{Z}^2$ must strike a balance between **capturing enough relevant spin-flip equations** and **remaining small enough to keep computations tractable**. Naively elongating $\mathcal{D} = 13^n 1$ for $n \gtrsim 4$ quickly exceeds the computational limits of §1-3, even on computer clusters. **Idea:** probe the correlators captured by $\mathcal{D} = 13^n 1$ through a collection of pairs of 131 diamonds separated by distance $\{0, 1, \ldots, n\}$.



We can see from the table that for $n \geq 4$, the # of independent spins in the collection of split diamonds with separations up to n increases by exactly 19 per unit increase in n.

Why does it work? Initially, the exact formulation of the positive semi-definiteness conditions is $\sum_{i,j} t^{S_i(B_1)} t^{S_j(B_2)} \langle \underline{s}_{S_i(B_1)} \underline{s}_{S_j(B_2)} \rangle \geq 0$ $\forall t^{A_1}, t^{A_2} \in \mathbb{R}$ for some $B_1, B_2 \subseteq \mathcal{D}$, which can be written as $\vec{t}^{tr} M \vec{t} \geq 0$ for some matrix M which must then satisfy $M \succeq 0$, but as this must be true for all $\vec{t}$, it must also be true for all $\vec{t}$ such that $t_i = t^{S_i(B)}$ vanishes for all $S_i(B)$ that does not fit inside the union of two 131 diamonds.

n	2	3	4	5	6	7	8	9	10	11
# of ind. $\langle s \rangle$	8	20	42	61	80	99	118	137	156	175